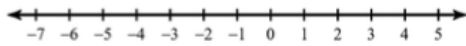


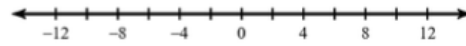
Solving Absolute Value Equations and Inequalities

Solve each inequality and graph its solution.

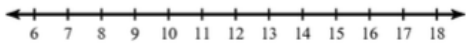
17) $|n| \geq 2$



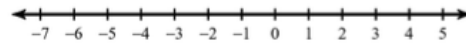
18) $|m| \leq 10$



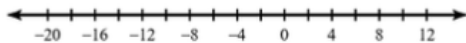
19) $|x - 8| \leq 1$



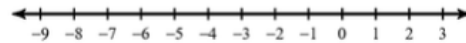
20) $|6b + 9| \leq 27$



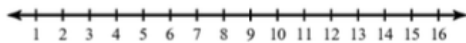
21) $8 + |a + 4| \geq 22$



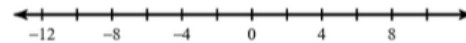
22) $|m + 4| + 2 \leq 5$



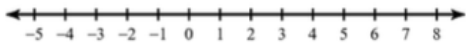
23) $5|9 - x| > 15$



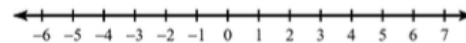
24) $|1 - 6b| + 2 \geq 51$



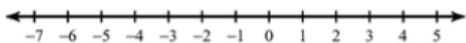
25) $8|2m - 5| - 7 < -63$



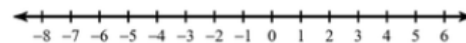
26) $-9|4n - 8| - 7 < 65$



27) $6|m| - 5 < 13$



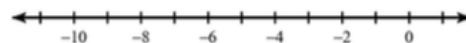
28) $|-2a| - 6 > 0$



29) $10 - 4|a + 6| < -30$



30) $8|7n + 7| - 4 \leq 52$





Practice questions 2.2.3

1 Find the value for each number.

a $|\pi|$

b $|-2|$

c $\left|\frac{3}{2}\right|$

d $|-\sqrt{2}|$

2 Simplify and work out the absolute value expressions.

a $|121 - 139|$

b $|3^2 - \sqrt{16}|$

c $\left|\frac{3 \times 4 - 2^2}{16 - 5^2}\right|$

d $|(3 - 12) + (25 - 34)|$

3 Solve each equation.

a $|x - 6| = 5$

b $|2x - 5| = 11$

c $\left|\frac{2x}{3}\right| = 4$

d $\left|\frac{x - 5}{6}\right| = 2$

4 Solve each inequality.

a $|x - 11| < 3$

b $|x + 3| \geq 3$

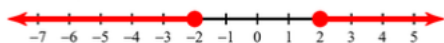
c $\left|\frac{x}{3}\right| \leq 5$

d $\left|\frac{x + 2}{4}\right| > 3$

Solutions:

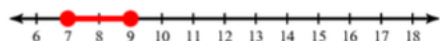
Answers

17) $|n| \geq 2$



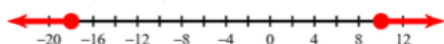
$n \geq 2$ or $n \leq -2$

19) $|x - 8| \leq 1$



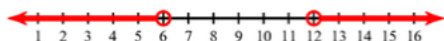
$7 \leq x \leq 9$

21) $8 + |a + 4| \geq 22$



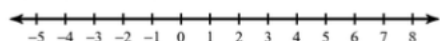
$a \geq 10$ or $a \leq -18$

23) $5|9 - x| > 15$



$x < 6$ or $x > 12$

25) $8|2m - 5| - 7 < -63$



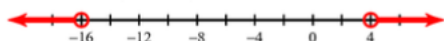
No solution.

27) $6|m| - 5 < 13$



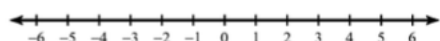
$-3 < m < 3$

29) $10 - 4|a + 6| < -30$



$a > 4$ or $a < -16$

31) $\frac{|5x - 7|}{10} \leq -3$



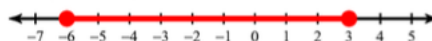
No solution.

18) $|m| \leq 10$



$-10 \leq m \leq 10$

20) $|6b + 9| \leq 27$



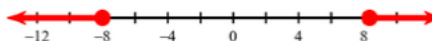
$-6 \leq b \leq 3$

22) $|m + 4| + 2 \leq 5$



$-7 \leq m \leq -1$

24) $|1 - 6b| + 2 \geq 51$



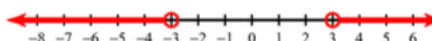
$b \leq -8$ or $b \geq \frac{25}{3}$

26) $-9|4n - 8| - 7 < 65$



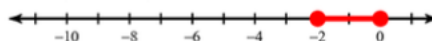
{ All real numbers. }

28) $|-2a| - 6 > 0$



$a < -3$ or $a > 3$

30) $8|7n + 7| - 4 \leq 52$



$-2 \leq n \leq 0$

32) $-5|1 - 6k| \leq 85$



{ All real numbers. }

Practice questions 2.2.3

1 a π

b 2

c $\frac{3}{2}$

d $\sqrt{2}$

2 a 18

b 5

c $\frac{8}{9}$

d 18

3 a $x = 1$ or $x = 11$

b $x = -3$ or $x = 8$

c $x = 6$ or $x = -6$

d $x = -7$ or $x = 17$

4 a $8 < x < 14$

b $x \geq 0$ or $x \leq -6$

c $-15 \leq x \leq 15$

d $x < -14$ or $x > 10$